

# CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

**PRE-MEDICAL : ENTHUSIAST COURSE [ PHASE-I ]**

**TEST # 07**

**DATE : 21-07-2019**

**Test Type : Minor**

**Test Pattern : NEET**

## INSTRUCTIONS

**Do not open this Test Booklet until you are asked to do so**

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

### Corporate Office

**ALLEN** CAREER INSTITUTE

"SANKALP", CP-6, Indra Vihar, Kota (Rajasthan)-324005

☎ +91-744-2757575 ☎ +91-744-2435003 ✉ info@allen.ac.in

### Rajkot Centres

#### Kalawad Road

"ISHAVASYAM", B/h. Crystal Mall, Opp./O.C. Quarters,  
Kalawad Road, Rajkot (Gujarat)-360005

☎ 0281-2562131 ☎ +91-85111-43783 ✉ rajkot@allen.ac.in

#### Limbudiwadi Centre

Opp. Landmark 2 Apartment, Limbudiwadi  
Main Road, Off. Kalawad Road, Rajkot (Gujarat)- 360007

☎ 0281-2450228

www.allen.ac.in

**Your Target is to secure Good Rank in NEET 2020**

**MINOR TEST # 07**

## Topics

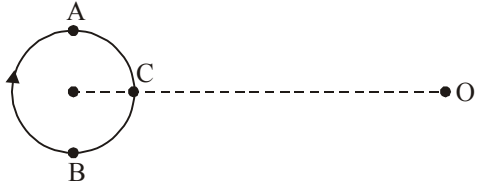
**DATE : 21 - 07 - 2019****PRE-MEDICAL ENTHUSIAST COURSE : PHASE - I – PATTERN : [NEET]**

|                  |                                                                                                                                                             |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PHYSICS</b>   | : Capacitors, Wave optics, String wave and Sound                                                                                                            |
| <b>CHEMISTRY</b> | : Chemistry in every day life, Polymers, Nitrogen and Oxygen containing organic compound, Equilibrium, S-Block, Nomenclature, Isomerism, GOC                |
| <b>BIOLOGY</b>   | : Genetic I (Till Linkage) Genetic II (Till Translation), Morphology, Plant anatomy, Immune system, Human disease (Bacterial and Viral), Geometrical system |

**SPACE FOR ROUGH WORK**

**HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS**

**BEWARE OF NEGATIVE MARKING**

- The displacement of a wave is represented by  
 $y = 0.25 \times 10^{-3} \sin (500 t - 0.05 x)$   
 where all the quantities are in their S.I. unit. The maximum particle velocity of the medium is .  
 (1)  $0.5 \text{ ms}^{-1}$  (2)  $0.75 \text{ ms}^{-1}$   
 (3)  $0.125 \text{ ms}^{-1}$  (4)  $0.150 \text{ ms}^{-1}$
- If the ratio of the amplitudes of two interfering beams be 2 : 3, the ratio of the minimum and maximum intensities of sound would be  
 (1) 4 : 9 (2)  $\sqrt{2} : \sqrt{3}$   
 (3) 1 : 5 (4) 1 : 25
- In the interference of waves from two sources of intensities  $I_0$  and  $4I_0$ , the intensity at a point where the phase difference is  $\pi$  is  
 (1)  $5I_0$  (2)  $3I_0$  (3)  $2I_0$  (4)  $I_0$
- The temperature at which the velocity of sound is double its velocity at  $0^\circ\text{C}$  is  
 (1)  $546^\circ\text{C}$  (2)  $819^\circ\text{C}$   
 (3)  $1092^\circ\text{C}$  (4)  $273^\circ\text{C}$
- Two waves represented by  $y = a \sin (\omega t - kx)$  and  $y = a \cos (\omega t - kx)$  are superposed. The resultant wave will have an amplitude  
 (1)  $a$  (2)  $\sqrt{2} a$  (3)  $2 a$  (4) 0
- A wave is represented by the equation  
 $y = (0.001 \text{ mm}) \sin [(50 \text{ s}^{-1})t + (2.0 \text{ m}^{-1})x]$ .  
 (1) The wave velocity = 100 m/s  
 (2) The wavelength = 2.0 m.  
 (3) The frequency =  $25/\pi$  Hz.  
 (4) The amplitude = 0.002 mm.
- Two sine waves travel in the same direction in a medium. The amplitude of each wave is A and the phase difference between the two waves is  $120^\circ$ . The resultant amplitude will be  
 (1) A (2) 2A (3) 4A (4)  $\sqrt{2}A$
- A cylindrical tube, open at both ends, has a fundamental frequency  $v$ . The tube is dipped vertically in water so that half of its length is inside the water. The new fundamental frequency is .  
 (1)  $v/4$  (2)  $v/2$  (3)  $v$  (4)  $2v$
- A small source of sound moves on a circle as shown in figure and an observer is sitting at O. Let  $v_1$ ,  $v_2$ ,  $v_3$  be the frequencies heard when the source is at A, B and C respectively.  
  
 (1)  $v_1 > v_2 > v_3$  (2)  $v_1 = v_2 > v_3$   
 (3)  $v_2 > v_3 > v_1$  (4)  $v_1 > v_3 > v_2$
- Two sources of waves are called coherent if  
 (1) Both have same amplitude of vibration  
 (2) Both produce waves of same wavelength  
 (3) Both produce waves of the same wavelength having constant phase difference  
 (4) Both produce waves having same velocity

SPACE FOR ROUGH WORK

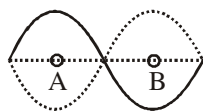
11. Two identical light waves, propagating in the same direction, have a phase difference  $\delta$ . After they superpose, the intensity of the resulting wave will be proportional to

(1)  $\cos \delta$  (2)  $\cos\left(\frac{\delta}{2}\right)$   
(3)  $\cos^2\left(\frac{\delta}{2}\right)$  (4)  $\cos^2 \delta$

12. For destructive interference to take place between two monochromatic light waves of wavelength  $\lambda$ , the path difference should be

(1)  $\frac{(2n-1)\lambda}{4}$  (2)  $2n\lambda$   
(3)  $\frac{(2n+1)\lambda}{2}$  (4)  $n\lambda$

13. In a standing wave particles at the position A and B, have a phase difference of



(1) 0 (2)  $\frac{\pi}{2}$  (3)  $\frac{5\pi}{6}$  (4)  $\pi$

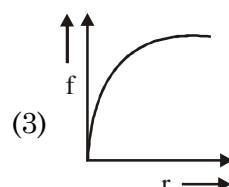
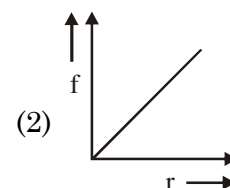
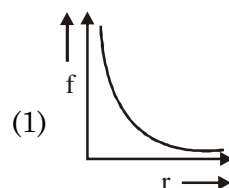
14. The change in phase if a wave is reflected from a rigid surface is

(1) Zero (2)  $\pi$  radian  
(3)  $2\pi$  radian (4)  $\frac{\pi}{2}$  radian

15. The number of compressions and rarefactions which pass an observer in 5 minutes, if the velocity of sound wave is 330 m/s and the wavelength of the wave is 30 cm, is

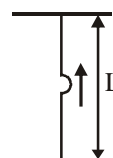
(1) 3300 (2)  $3.3 \times 10^4$   
(3)  $3.3 \times 10^5$  (4)  $3.3 \times 10^2$

16. Which of the following represent the variation of frequency (f) with radius of cross section of a stretched string ?



(4) None of these

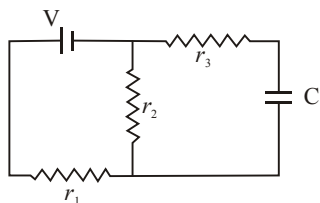
17. A thick uniform rope of length L is hanging from a rigid support. A transverse wave of wavelength  $\lambda_0$  is set up in the middle of the rope. The wavelength of the wave as it reaches the top most point is



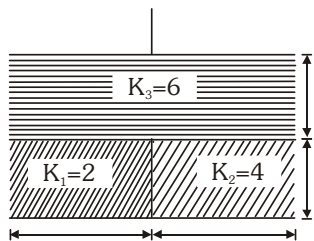
(1)  $2\lambda_0$  (2)  $\sqrt{2}\lambda_0$  (3)  $\frac{\lambda_0}{\sqrt{2}}$  (4)  $\lambda_0$

SPACE FOR ROUGH WORK

18. In the circuit shown in the figure, the final voltage drop across the capacitor C in steady state is



- (1)  $\frac{Vr_1}{r_1 + r_2}$  (2)  $\frac{Vr_2}{r_1 + r_2}$   
(3)  $\frac{V(r_1 + r_2)}{r_2}$  (4)  $\frac{V(r_1 + r_2)}{r_1 + r_2 + r_3}$
19. A parallel plate capacitor of capacitance C (without dielectrics) is filled by dielectric slabs as shown in figure. Then the new capacitance of the capacitor is



- (1) 3.9 C (2) 4 C  
(3) 2.4 C (4) 3 C
20. A capacitor of  $20 \mu\text{F}$  is charged to 500 V. Now battery is disconnected and uncharged capacitor of  $5 \mu\text{F}$  is connected in parallel. Potential difference across combination of capacitor is .....
- (1) 400 V (2) 300 V  
(3) 200 V (4) 100 V

21. A capacitor with air as medium between plate is charged to voltage V. Now battery is disconnected and distance between plate is increased then potential difference between plates.....

(1) do not change (2) become zero  
(3) increases (4) decreases

22. If Q is the charge on the plates of a capacitor of capacitance C, V the potential difference between the plates, A the area of each plate and d the distance between the plates, the force of attraction between the plates is

(1)  $\frac{1}{2} \left( \frac{Q^2}{\epsilon_0 A} \right)$  (2)  $\frac{1}{2} \left( \frac{CV^2}{d^2} \right)$

(3)  $\frac{1}{2} \left( \frac{CV^2}{A\epsilon_0} \right)$  (4)  $\frac{1}{4} \left( \frac{Q^2}{\pi\epsilon_0 d^2} \right)$

23. The area of every plate shown in the figure is A and the separation between the successive plates is d. What is the capacitance between points a and b ?



(1)  $\epsilon_0 A/d$  (2)  $2\epsilon_0 A/d$   
(3)  $3\epsilon_0 A/d$  (4)  $4\epsilon_0 A/d$

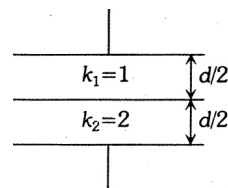
24. Two capacitor having value  $C_1$  and  $C_2$  are connected in parallel. This combination is given charge Q. Ratio of charges on  $C_1$  and  $C_2$  is .....

(1)  $C_1 C_2$  (2)  $\frac{C_2}{C_1}$  (3)  $C_1 + C_2$  (4)  $\frac{C_1}{C_2}$

SPACE FOR ROUGH WORK

25. How many capacitors of  $8\mu\text{F}$ ,  $250\text{V}$  rating are connected in which way so that equivalent capacitance is  $16\mu\text{F}$  and  $1000\text{V}$  rating obtained?  
(1) 4 (2) 32 (3) 8 (4) 3
26. The energy required to charge a parallel plate condenser of plate separation  $d$  and plate area of cross-section  $A$  such that the uniform electric field between the plates is  $E$ , is  
(1)  $\epsilon_0 E^2 Ad$  (2)  $\frac{1}{2}\epsilon_0 E^2 Ad$   
(3)  $\frac{1}{2}\epsilon_0 E^2 / A.d$  (4)  $\epsilon_0 E^2 / Ad$
27. Eight small drops, each of radius  $r$  and having same charge  $q$  are combined to form a big drop. The ratio between the potentials of the bigger drop and the smaller drop is  
(1) 8 : 1 (2) 4 : 1  
(3) 2 : 1 (4) 1 : 8
28. A parallel plate capacitor is charged to a potential difference of 50 volts. It is then discharged through a resistance for 2 seconds and its potential drops by 10 volts. Calculate the fraction of energy stored in the capacitance  
(1) 0.14 (2) 0.25  
(3) 0.50 (4) 0.64
29. The capacity of a parallel plate capacitor increases with the  
(1) Decrease of its area  
(2) Increase of its distance  
(3) Increase of its area  
(4) None of the above

30. Two parallel plate of area  $A$  are separated by two different dielectrics as shown in figure. The net capacitance is



- (1)  $\frac{4\epsilon_0 A}{3d}$  (2)  $\frac{3\epsilon_0 A}{4d}$  (3)  $\frac{2\epsilon_0 A}{d}$  (4)  $\frac{\epsilon_0 A}{d}$
31. There is an air filled  $1\text{pF}$  parallel plate capacitor. When the plate separation is doubled and the space is filled with wax, the capacitance increases to  $2\text{pF}$ . The dielectric constant of wax is  
(1) 2 (2) 4  
(3) 6 (4) 8
32. The capacity and the energy stored in a parallel plate condenser with air between its plates are respectively  $C_0$  and  $W_0$ . If the air is replaced by glass (dielectric constant = 5) between the plates, the capacity of the plates and the energy stored in it will respectively be  
(1)  $5C_0, 5W_0$  (2)  $5C_0, \frac{W_0}{5}$   
(3)  $\frac{C_0}{5}, 5W_0$  (4)  $\frac{C_0}{5}, \frac{W_0}{5}$
33. A parallel plate capacitor is made by stacking  $n$  equally spaced plates connected alternately. If the capacitance between any two plates is  $C$  then the resultant capacitance is  
(1)  $C$  (2)  $nC$   
(3)  $(n-1)C$  (4)  $(n+1)C$

SPACE FOR ROUGH WORK

34. Three capacitors each of capacitance  $C$  and of breakdown voltage  $V$  are joined in series. The capacitance and breakdown voltage of the combination will be

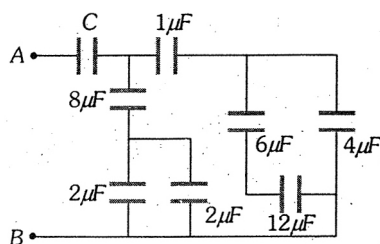
- (1)  $\frac{C}{3}, \frac{V}{3}$  (2)  $3C, \frac{V}{3}$   
(3)  $\frac{C}{3}, 3V$  (4)  $3C, 3V$

35. Three capacitors of capacitances  $3\mu\text{F}$ ,  $9\mu\text{F}$  and  $18\mu\text{F}$  are connected once in series and another time in parallel. The ratio of equivalent capacitance in the two cases

$\left(\frac{C_s}{C_p}\right)$  will be

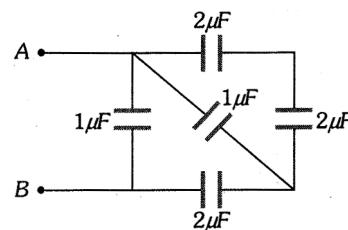
- (1) 1 : 15 (2) 15 : 1  
(3) 1 : 1 (4) 1 : 3

36. In the following circuit, the resultant capacitance between A and B is  $1\mu\text{F}$ . Then value of  $C$  is



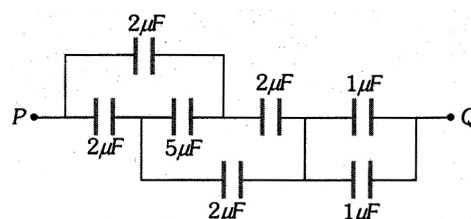
- (1)  $\frac{32}{11}\mu\text{F}$  (2)  $\frac{11}{32}\mu\text{F}$   
(3)  $\frac{23}{32}\mu\text{F}$  (4)  $\frac{32}{23}\mu\text{F}$

37. The total capacity of the system of capacitors shown in the adjoining figure between the points A and B is



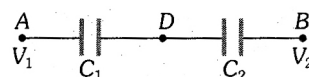
- (1)  $1\mu\text{F}$  (2)  $2\mu\text{F}$   
(3)  $3\mu\text{F}$  (4)  $4\mu\text{F}$

38. The effective capacitance between the points P and Q of the arrangement shown in the figure is



- (1)  $\frac{1}{2}\mu\text{F}$  (2)  $1\mu\text{F}$   
(3)  $2\mu\text{F}$  (4)  $1.33\mu\text{F}$

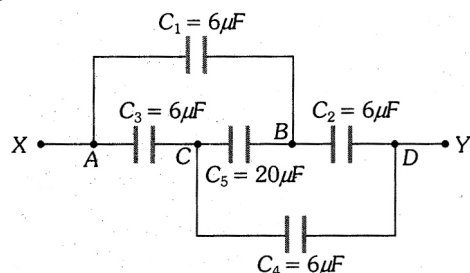
39. Two condensers  $C_1$  and  $C_2$  in a circuit are joined as shown in figure. The potential of point A is  $V_1$  and that of B is  $V_2$ . The potential of point D will be



- (1)  $\frac{1}{2}(V_1 + V_2)$  (2)  $\frac{C_2 V_1 + C_1 V_2}{C_1 + C_2}$   
(3)  $\frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$  (4)  $\frac{C_2 V_1 - C_1 V_2}{C_1 + C_2}$

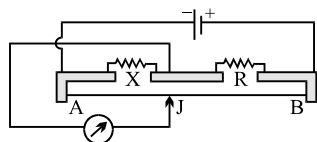
SPACE FOR ROUGH WORK

40. What is the effective capacitance between points X and Y



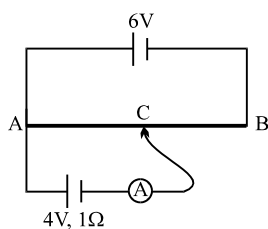
- (1)  $24\mu\text{F}$  (2)  $18\mu\text{F}$  (3)  $12\mu\text{F}$  (4)  $6\mu\text{F}$

41. The figure shows a metre-bridge circuit, with  $AB = 100\text{ cm}$ ,  $X = 12\Omega$  and  $R = 18\Omega$ , and the jockey  $J$  in the position of balance. If  $R$  is now made  $8\Omega$ , through what distance will  $J$  have to be moved to obtain balance?



- (1) 10 cm (2) 20 cm  
(3) 30 cm (4) 40 cm

42. A 6 V battery of negligible internal resistance is connected across a uniform



wire of length 1 m. The positive terminal of another battery of emf 4V and internal resistance  $1\Omega$  is joined to the point A as shown in figure. The ammeter shows zero deflection when the jockey touches the wire at the point C. The AC is equal to

- (1)  $2/3\text{ m}$  (2)  $1/3\text{ m}$  (3)  $3/5\text{ m}$  (4)  $1/2\text{ m}$

43. A galvanometer has a resistance of  $20\Omega$  and reads full-scale when 0.2 V is applied across it. To convert it into a 10 A ammeter, the galvanometer coil should have a

- (1)  $0.01\Omega$  resistor connected across it  
(2)  $0.02\Omega$  resistor connected across it  
(3)  $200\Omega$  resistor connected in series with it  
(4)  $2000\Omega$  resistor connected in series with it

44. A galvanometer coil has a resistance  $90\Omega$  and full scale deflection current 10 mA. A  $910\Omega$  resistance is connected in series with the galvanometer to make a voltmeter. If the least count of the voltmeter is 0.1V, the number of divisions on its scale is

- (1) 90  
(2) 91  
(3) 100  
(4) none

45. A potentiometer wire has length 10 m and resistance  $10\Omega$ . It is connected to a battery of EMF 11 volt and internal resistance  $1\Omega$ , then the potential gradient in the wire is

- (1) 10 V/m  
(2) 1 V/m  
(3) 0.1 V/m  
(4) none

SPACE FOR ROUGH WORK

46. First ionisation energy of alkali metals is very low but second ionisation energy is very high because  
 (1) alkali metals acquire noble gas configuration after losing one electron  
 (2) a large amount of energy is required to remove electron from a cation  
 (3) alkali metals can form only univalent ions  
 (4) first group elements can lose only one electron.
47. Which of the following alkali metals when burnt in air forms a mixture of oxide as well as nitride?  
 (1) K (2) Na  
 (3) Li (4) Cs
48. On reaction with dihydrogen the alkali metals  
 (1) form hydrides which are ionic solids with high melting points  
 (2) form hydrides which are molecular solids with low melting points  
 (3) form hydrides which are ionic solids with low melting points  
 (4) form hydrides which are non-stoichiometric.
49. Lithium is the strongest reducing agent though it has highest ionisation energy in its group. Which of the following factor is responsible for making Li the strongest reducing agent?  
 (1) Large heat of atomisation  
 (2) Smaller size  
 (3) Large sublimation energy  
 (4) Large amount of hydration enthalpy
50. The alkali metals dissolve in ammonia to give a deep blue solution which is conducting in nature.  

$$M + (x+y) NH_3 \rightarrow [M(NH_3)_x]^{2+} + 2[e(NH_3)_y]^-$$
 Which of the following is not true about the solution of alkali metals in liquid ammonia?  
 (1) The blue colour is due to ammoniated electron  
 (2) The solution is paramagnetic.  
 (3) The blue colour changes to brown on standing  
 (4) In concentrated solution blue colour changes to bronze and becomes diamagnetic.
51. Gypsum is added to portland cement to  
 (1) fasten the process of setting  
 (2) slow down the process of setting  
 (3) improve the colour of the cement  
 (4) increase the melting point of cement.
52. Match the column I with column II and mark the appropriate choice.
- | Column I                                            | Column II               |
|-----------------------------------------------------|-------------------------|
| (1) Liquid $\rightleftharpoons$ Vapour              | (i) Saturated solution  |
| (2) Solid $\rightleftharpoons$ Liquid               | (ii) Boiling point      |
| (3) Solid $\rightleftharpoons$ Vapour               | (iii) Sublimation point |
| (4) Solute(s) $\rightleftharpoons$ Solute(solution) | (iv) Melting point      |
- (1) (1)  $\rightarrow$  (i), (2)  $\rightarrow$  (iii), (3)  $\rightarrow$  (ii), (4)  $\rightarrow$  (iv)  
 (2) (1)  $\rightarrow$  (ii), (2)  $\rightarrow$  (iv), (3)  $\rightarrow$  (iii), (4)  $\rightarrow$  (i)  
 (3) (1)  $\rightarrow$  (iv), (2)  $\rightarrow$  (ii), (3)  $\rightarrow$  (i), (4)  $\rightarrow$  (iii)  
 (4) (1)  $\rightarrow$  (iii), (2)  $\rightarrow$  (iv), (3)  $\rightarrow$  (ii), (4)  $\rightarrow$  (i)

SPACE FOR ROUGH WORK

53. If the equilibrium constant for the given reaction is 0.25

$\text{NO} \rightleftharpoons \frac{1}{2}\text{N}_2 + \frac{1}{2}\text{O}_2$ , then the equilibrium constant for the reaction

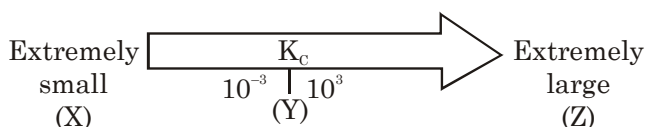
$\frac{1}{2}\text{N}_2 + \frac{1}{2}\text{O}_2 \rightleftharpoons \text{NO}$  will be

- (1) 1 (2) 2  
(3) 3 (4) 4

54. Which of the following is an example of homogeneous equilibrium?

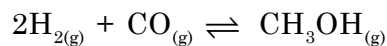
- (1)  $2\text{SO}_{2(g)} + \text{O}_{2(g)} \rightleftharpoons 2\text{SO}_{3(g)}$   
(2)  $\text{C}_{(s)} + \text{H}_2\text{O}_{(g)} \rightleftharpoons \text{CO}_{(g)} + \text{H}_{2(g)}$   
(3)  $\text{CaCO}_{3(s)} \rightleftharpoons \text{CaO}_{(s)} + \text{CO}_{2(g)}$   
(4)  $\text{NH}_4\text{HS}_{(s)} \rightleftharpoons \text{NH}_{3(g)} + \text{H}_2\text{S}_{(g)}$

55. Study the figure given below and mark the correct statement about  $K_c$  and dependence of extent of reaction on it.



|     | X                               | Y                                         | Z                                         |
|-----|---------------------------------|-------------------------------------------|-------------------------------------------|
| (1) | Reaction does not occur         | Reaction proceeds to completion           | Reaction does not occur                   |
| (2) | Reaction completes              | Reaction does not occur                   | Reactants and products are at equilibrium |
| (3) | Reaction hardly occurs          | Reactants and products are at equilibrium | Reaction proceeds to completion           |
| (4) | Reaction proceeds to completion | Reactants and products are at equilibrium | Reaction hardly occurs                    |

56. Consider the equilibrium set up:

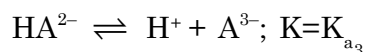
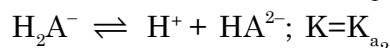
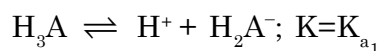


What will be the effect of the following on the equilibrium of the reaction?

- (i) Addition of  $\text{H}_2$   
(ii) Addition of  $\text{CH}_3\text{OH}$   
(iii) Removal of  $\text{CO}$   
(iv) Removal of  $\text{CH}_3\text{OH}$

|     | (i)                | (ii)               | (iii)              | (iv)               |
|-----|--------------------|--------------------|--------------------|--------------------|
| (1) | Forward direction  | Backward direction | Backward direction | Forward direction  |
| (2) | Backward direction | Backward direction | Forward direction  | Forward direction  |
| (3) | Forward direction  | Forward direction  | Backward direction | Backward direction |
| (4) | Backward direction | Forward direction  | Forward direction  | Backward direction |

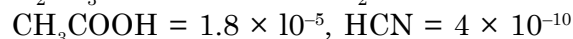
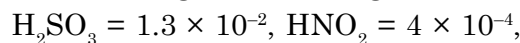
57. For a polybasic acid, the dissociation constants have a different values for each step, e.g.,



What is the observed trend of dissociation constants in successive stages?

- (1)  $K_{a_1} > K_{a_2} > K_{a_3}$  (2)  $K_{a_1} = K_{a_2} = K_{a_3}$   
(3)  $K_{a_1} < K_{a_2} < K_{a_3}$  (4)  $K_{a_1} = K_{a_2} + K_{a_3}$

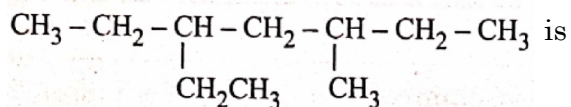
58. Given below are the dissociation constant values of few acids. Arrange them in order of increasing acidic strength.



- (1)  $\text{HCN} < \text{CH}_3\text{COOH} < \text{HNO}_2 < \text{H}_2\text{SO}_3$   
(2)  $\text{CH}_3\text{COOH} < \text{HNO}_2 < \text{HCN} < \text{H}_2\text{SO}_3$   
(3)  $\text{CH}_3\text{COOH} < \text{HCN} < \text{H}_2\text{SO}_3 < \text{HNO}_2$   
(4)  $\text{HNO}_2 < \text{H}_2\text{SO}_3 < \text{CH}_3\text{COOH} < \text{HCN}$

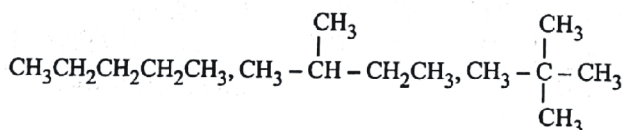
SPACE FOR ROUGH WORK

59. Correct name for the given compound,



- (1) 3-ethyl-5-methylheptane
- (2) 5-ethyl-3-methylheptane
- (3) 1,1-diethyl-3-methylpentane
- (4) 3-methyl-5,5-diethylpentane.

60. The type of isomerism shown by the following compounds is



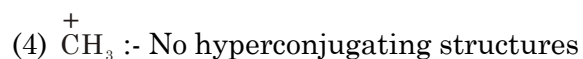
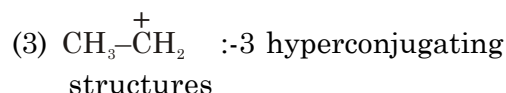
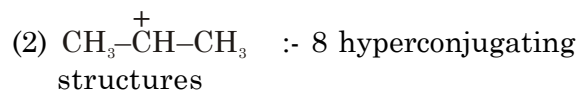
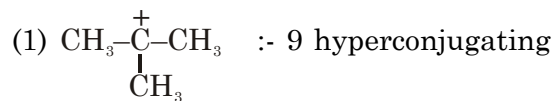
- (1) position isomerism
- (2) metamerism
- (3) ring-chain isomerism
- (4) chain isomerism.

61. Given below are the structures of few compounds with molecular formula  $\text{C}_4\text{H}_{10}\text{O}$ . Select metamers from these structures.

- (i)  $\text{CH}_3 - \text{O} - \text{CH}_2\text{CH}_2\text{CH}_3$
- (ii)  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
- (iii)  $\text{CH}_3 - \text{CH}_2 - \text{O} - \text{CH}_2 - \text{CH}_3$
- (iv)  $\text{CH}_3 - \underset{\text{OH}}{\text{CH}} - \text{CH}_2 - \text{CH}_3$

- (1) (i) and (ii)
- (2) (ii) and (iii)
- (3) (i) and (iii)
- (4) (ii) and (iv)

62. The number of hyperconjugating structures shown by the carbocations are given below. Which one is not correctly matched ?



63. Which alkane is produced when sodium salt of butanoic acid is heated with soda lime?

- (1)  $\text{CH}_3\text{CH}_3$
- (2)  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
- (3)  $\text{CH}_4$
- (4)  $\text{CH}_3\text{CH}_2\text{CH}_3$

64. hydrocarbon which is liquid at room temperature is

- (1) pentane
- (2) butane
- (3) propane
- (4) ethane

65. Chlorination of alkanes is a photochemical process. It is initiated by the process of

- (1) heterolysis
- (2) homolysis
- (3) pyrolysis
- (4) hydrolysis.

SPACE FOR ROUGH WORK

66. Which step is chain propagation step in the following mechanism?

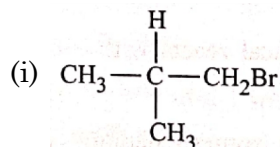
- (i)  $\text{Cl}_2 \xrightarrow{h\nu} \text{Cl}^\bullet + \text{Cl}^\bullet$   
 (ii)  $\dot{\text{Cl}} + \text{CH}_4 \longrightarrow \dot{\text{C}}\text{H}_3 + \text{HCl}$   
 (iii)  $\text{Cl}^\bullet + \text{Cl}^\bullet \longrightarrow \text{Cl}_2$   
 (iv)  $\text{CH}_3 + \dot{\text{Cl}} \longrightarrow \text{CH}_3\text{Cl}$

(1) (i) (2) (ii) (3) (iii) (4) (iv)

67. Which of the following products is formed when n-heptane is passed over  $(\text{Al}_2\text{O}_3 + \text{Cr}_2\text{O}_3)$  catalyst at 773 K?

- (1) Benzene (2) Toluene  
 (3) Polyheptane (4) Cycloheptane

68. Arrange the following alkyl halides in decreasing order of the rate of  $\beta$ -elimination reaction with alcoholic KOH.



- (ii)  $\text{CH}_3 - \text{CH}_2 - \text{Br}$   
 (iii)  $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{Br}$

- (1) i > ii > iii (2) iii > ii > i  
 (3) ii > iii > i (4) i > iii > ii

69. Which of the following reactions does not show the correct products of the reaction?

- (1)  $\text{CH}_3 - \text{CH} = \text{CH}_2 \xrightarrow[\text{peroxide}]{\text{HBr}} \text{CH}_3 - \text{CH}_2 - \text{CH}_2\text{Br}$   
 (2)  $\text{CH}_3 - \text{CH} = \text{CH}_2 \xrightarrow[\text{peroxide}]{\text{HCl}} \text{CH}_3 - \text{CH}_2 - \text{CH}_2\text{Cl}$   
 (3)  $\text{CH}_3 - \text{CH} = \text{CH}_2 \xrightarrow{\text{HBr}} \text{CH}_3 - \overset{\text{Br}}{\text{CH}} - \text{CH}_3$   
 (4)  $\text{CH}_3 - \text{CH} = \text{CH}_2 \xrightarrow{\text{HCl}} \text{CH}_3 - \overset{\text{Cl}}{\text{CH}} - \text{CH}_3$

70. What happens when calcium carbide is treated with water?

- (1) Ethane is formed.  
 (2) Methane and ethane are formed.  
 (3) Ethyne is formed.  
 (4) Ethene and ethyne are formed.

71. Fill in the blanks with appropriate words. Benzene has a planar structure. All carbon atoms in benzene are \_\_I\_\_ hybridised. The ring structure of benzene was proposed by \_\_II\_\_. It shows \_\_III\_\_ substitution reactions. It reacts with \_\_IV\_\_ in presence of aluminium chloride to form acetophenone.

|     | I             | II        | III           | IV              |
|-----|---------------|-----------|---------------|-----------------|
| (1) | $\text{sp}^2$ | Kekule    | electrophilic | acetyl chloride |
| (2) | $\text{sp}$   | Dewar     | nucleophilic  | chloromethane   |
| (3) | $\text{sp}^3$ | Ladenberg | electrophilic | chloroethane    |
| (4) | $\text{sp}^2$ | Baeyer    | nucleophilic  | methyl bromide  |

72. Nitration and chlorination of benzene are  
 (1) nucleophilic and electrophilic substitution respectively

- (2) electrophilic and nucleophilic substitution respectively  
 (3) electrophilic substitution in both the reactions  
 (4) nucleophilic substitution in both the reactions.

73. An alkene  $\text{CH}_3\text{CH} = \text{CH}_2$  is treated with  $\text{B}_2\text{H}_6$  in presence of  $\text{H}_2\text{O}_2$ . The final product formed is

- (1)  $\text{CH}_3\text{CH}_2\text{CHO}$   
 (2)  $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$   
 (3)  $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$   
 (4)  $(\text{CH}_3\text{CH}_2\text{CH}_2)_3\text{B}$

SPACE FOR ROUGH WORK

74. Tertiary butyl alcohol can be prepared by the reaction of  
 (1) acetaldehyde and ethyl magnesium iodide  
 (2) acetone and methyl magnesium iodide  
 (3) formaldehyde and propyl magnesium iodide  
 (4) butanone and methyl magnesium iodide.
75. Which of the following alcohols reacts most readily with Lucas reagent?  
 (1)  $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$   
 (2)  $\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\text{CH}_3$   
 (3)  $\text{CH}_3-\underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}}-\text{OH}$   
 (4)  $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2\text{OH}$
76. Which of the following alcohols will give the most stable carbocation during dehydration?  
 (1) 2-Methyl-1-propanol  
 (2) 2-Methyl-2-propanol  
 (3) 1-Butanol  
 (4) 2-Butanol
77.  $\text{C}_3\text{H}_9\text{N}$  cannot represent  
 (1)  $1^\circ$  amine  
 (2) quaternary ammonium salt  
 (3)  $2^\circ$  amine  
 (4)  $3^\circ$  amine

78. Aldehydes that do not undergo aldol condensation are  
 (1) propanal  
 (2) trichloroethanal  
 (3) 2-phenylethanal  
 (4) ethanal  
 (5) benzaldehyde  
 (1) 3 and 4 only      (2) 2 and 5 only  
 (3) 1,2 and 3 only      (4) 2,3 and 5 only
79. Match the column I with column II and mark the appropriate choice.

| Column I |                      | Column II |                                      |
|----------|----------------------|-----------|--------------------------------------|
| (A)      | Clemmensen reduction | (i)       | Conc. KOH                            |
| (B)      | Rosenmund reduction  | (ii)      | Zn/Hg + conc. HCl                    |
| (C)      | Iodoform reaction    | (iii)     | $\text{H}_2/\text{Pd}-\text{BaSO}_4$ |
| (D)      | Cannizzaro reaction  | (iv)      | $\text{NaOH} + \text{I}_2$           |

- (1) (A)→(i), (B)→(iii), (C)→(ii), (D)→(iv)  
 (2) (A)→(iii), (B)→(iv), (C)→(i), (D)→(ii)  
 (3) (A)→(ii), (B)→(iii), (C)→(iv), (D)→(i)  
 (4) (A)→(iv), (B)→(i), (C)→(ii), (D)→(iii)
80. Reduction of aromatic nitro compounds using Sn and HCl gives  
 (1) aromatic primary amines  
 (2) aromatic secondary amines  
 (3) aromatic tertiary amines .  
 (4) aromatic amides.
81. Identify 'Z' in the sequence :  
 $\text{C}_6\text{H}_5\text{NH}_2 \xrightarrow[273\text{K}]{\text{NaNO}_2+\text{HCl}} \text{X} \xrightarrow{\text{CuCN}} \text{Y} \xrightarrow[\text{Boil}]{\text{H}^+/\text{H}_2\text{O}} \text{Z}$   
 (1)  $\text{C}_6\text{H}_5\text{CN}$   
 (2)  $\text{C}_6\text{H}_5\text{CONH}_2$   
 (3)  $\text{C}_6\text{H}_5\text{COOH}$   
 (4)  $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$

SPACE FOR ROUGH WORK

82. Match the column I with column II and mark the appropriate choice.

| Column I |                                                                            | Column II |                            |
|----------|----------------------------------------------------------------------------|-----------|----------------------------|
| (A)      | $\text{CH}_3\text{NH}_2 + \text{CHCl}_3 + \text{KOH} \xrightarrow{\Delta}$ | (i)       | $\text{CH}_3\text{NH}_2$   |
| (B)      | $\text{CH}_3\text{CONH}_2 + \text{Br}_2 + \text{KOH} \longrightarrow$      | (ii)      | $\text{CH}_3\text{OH}$     |
| (C)      | $\text{CH}_3\text{NH}_2 + \text{NaNO}_2 + \text{HCl} \longrightarrow$      | (iii)     | $\text{CH}_3\text{NHCH}_3$ |
| (D)      | $\text{CH}_3\text{NC} + 4\text{H} \xrightarrow{\text{Pt}}$                 | (iv)      | $\text{CH}_3\text{NC}$     |

- (1) (1)→(i), (2)→(ii), (3)→(iii), (4)→(iv)  
 (2) (1)→(ii), (2)→(iii), (3)→(iv), (4)→(i)  
 (3) (1)→(iv), (2)→(i), (3)→(ii), (4)→(iii)  
 (4) (1)→(iii), (2)→(iv), (3)→(i), (4)→(ii)

83. In vulcanization of rubber

- (1) sulphur reacts to form a new compound  
 (2) sulphur cross-links are introduced  
 (3) sulphur forms a very thin protective layer over rubber  
 (4) all statements are correct.

84. Identify the type of polymer.

- (i) —A—A—A—A—A—  
 (ii) —A—B—B—A—A—A—B—A—  
 (1) (i) Homopolymer, (ii) Copolymer  
 (2) (i) Natural polymer,  
 (ii) Synthetic polymer  
 (3) (i) Linear polymer,  
 (ii) Branched polymer  
 (4) (i) Fibre, (ii) Elastomer

85. Which of the following is a homopolymer?

- (1) Bakelite (2) Nylon 6,6  
 (3) Neoprene (4) Buna-S

86. Arrange the following polymers in an increasing order of intermolecular forces; fibre, plastic, elastomer.

- (1) Elastomer < Fibre < Plastic  
 (2) Elastomer < Plastic < Fibre  
 (3) Plastic < Elastomer < Fibre  
 (4) Fibre < Elastomer < Plastic

87. Which of the following is not true for thermoplastic polymers?

- (1) Thermoplastics are linear polymers.  
 (2) They soften and melt on heating.  
 (3) Molten polymer can be remoulded into any shape.  
 (4) They have cross-linkages which break on heating.

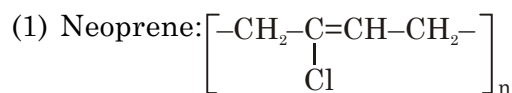
88. Low density polythene (LDP) is used in the insulation of electricity carrying wires and manufacture of flexible pipes and squeeze bottles because

- (1) it is tough, hard and rigid  
 (2) it is chemically inert, tough, flexible and poor conductor of electricity  
 (3) it is very tough, good conductor of electricity and flexible  
 (4) it is chemically inert, very soft, water absorbent and poor conductor of heat.

89. Composition of Ziegler-Natta catalyst is

- (1)  $(\text{Et})_3\text{Al} \cdot \text{TiCl}_2$  (2)  $(\text{Me})_3\text{Al} \cdot \text{TiCl}_2$   
 (3)  $(\text{Et})_3\text{Al} \cdot \text{TiCl}_4$  (4)  $(\text{Et})_3\text{Al} \cdot \text{PtCl}_4$

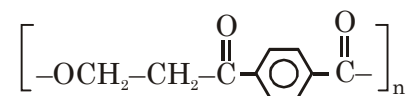
90. Which of the following is not correctly matched ?



- (2) Nylon-6,6:



- (3) Terylene:



- (4) Teflon :  $\left[ \text{CF}_2\text{—CF}_2 \right]_n$

SPACE FOR ROUGH WORK

- 91.** Which of the following sequence is found in Rous-Sarcoma virus?  
 (1) DNA → DNA → Protein  
 (2) RNA → DNA → RNA → Protein  
 (3) RNA → RNA → Protein  
 (4) DNA → RNA → Protein
- 92.** Genetic code is called degenerative because:  
 (1) Amino acid codes for different codons  
 (2) Amino acid can be coded by more than one codon  
 (3) One tRNA recognizes one amino acid  
 (4) One type of mRNA synthesizes many types of enzymes
- 93.** Which of the following character is not applicable to tRNA ?  
 (1) It is the smallest of the RNAs  
 (2) It acts as an adaptor for amino acid  
 (3) It has a clover leaf like structure  
 (4) It is the largest of the RNAs
- 94.** Considering that we have four nucleotides A, G, C, T, the number of base substitutions that can occur in the amino acid codons are:  
 (1) 549      (2) 535      (3) 261      (4) 264
- 95.** Which site of a mRNA molecule hydrogen binds to a rRNA molecule?  
 (1) Codon  
 (2) Anticodon  
 (3) 5' end of a mRNA molecule  
 (4) 3' end of a mRNA molecule
- 96.** Correct sequence of the transfer of genetic information during formation of a polypeptide is:  
 (1) DNA, tRNA, rRNA and mRNA  
 (2) mRNA, rRNA, DNA and amino acid  
 (3) rRNA, DNA, mRNA and tRNA  
 (4) DNA, mRNA, tRNA and amino acid
- 97.** If only two types of nucleotides are found in living world, genetic code is:  
 (1) doublet                      (2) Triplet  
 (3) Quadruplet                  (4) Quintuplet
- 98.** The genetic code was discovered by:  
 (1) Nirenberg and Matthaei  
 (2) Nirenberg and Holley  
 (3) Kornberg  
 (4) Holley, Nirenberg and Khorana
- 99.** The first mRNA synthesized for the assignment of codon was a:  
 (1) Polyadenylate  
 (2) Polycytidylate  
 (3) Polyguanylate  
 (4) Polyuridylate
- 100.** Which of the following statements are correct regarding transcription?  
 (i) The mRNA provides the template, tRNA brings amino acids and reads the genetic code, and rRNAs plays structural and catalytic role during translation  
 (ii) There is single DNA-dependent RNA polymerase that catalyses transcription of all types of RNA in bacteria.  
 (iii) RNA polymerase able to catalyse all the three steps of transcription i.e., initiation, elongation and termination  
 (iv) The RNA polymerase is only capable of catalysing the process of elongation.  
 (v) Many times the translation can begin much before the mRNA is fully transcribed in bacteria, so transcription and translation can be coupled in bacteria  
 (vi) In eukaryotes, there are at least three types of RNA polymerase in the nucleus  
 (vii) In eukaryotes, the primary transcripts called hnRNA contain both the exons and the introns and are non-functional  
 (1) All except (iii) and (iv)  
 (2) (i), (ii) and (iii)  
 (3) (ii), (iii), (v), (vi) and (vii)  
 (4) All except (iii) are correct

**101.** Which of the following statements are not correct?

- (i) The secondary structure of tRNA has been depicted that looks like a clover-leaf.
- (ii) In actual structure, the tRNA is a compact molecule which looks like inverted L.
- (iii) There are two sites in the large subunit of ribosome, for subsequent amino acids so that formation of a peptide bond occurs
- (vi) The ribosome also acts as a catalyst (23S rRNA in bacteria is the enzyme-ribozyme) for the formation of peptide bond.
- (v) A translational unit in mRNA have start codon (AUG), stop codon, codes for a polypeptide and untranslated regions (UTR).
- (vi) Untranslated regions (UTR) present at both 5'-end (before start codon) and at 3'-end (after stop codon) of translational unit in mRNA.

- (1) (v) and (vi)
- (2) (iii) and (iv)
- (3) Only (vi)
- (4) None of these

**102.** If the sequence of nitrogen bases of the coding strand of DNA in a transcription unit is 5'-A T G A A T G-3', the sequence of bases in its RNA transcript would be:

- (1) 5'-A U G A A U G-3'
- (2) 5'-U A C U U A C-3'
- (3) 5'-C A U U C A U-3'
- (4) 5'-G U A A G U A-3'

**103.** To initiate translation, the mRNA first binds to:

- (1) The smaller ribosomal sub-unit
- (2) The larger ribosomal sub-unit
- (3) The whole ribosome
- (4) No such specificity exists

**104. Assertion:-** The RNA polymerase associates transiently with initiation-factor ( $\sigma$ ) and termination-factor ( $\rho$ ) to initiate and terminate the transcription, respectively.

**Reason:-** Association with these factors alter the specificity of the RNA polymerase to either initiate or terminate

- (1) If both Assertion & Reason are true & the reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are true but reason is not a correct explanation of the Assertion.
- (3) If Assertion is true but the Reason is false
- (4) If both Assertion & Reason are false

**105. Assertion:-** The split-gene arrangements represent probably an ancient feature of the genome.

**Reason:-** The presence of introns is reminiscent of antiquity, and the process of splicing represents the dominance of RNA-world

- (1) If both Assertion & Reason are true & the reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are true but reason is not a correct explanation of the Assertion.
- (3) If Assertion is true but the Reason is false
- (4) If both Assertion & Reason are false

**106.** The codon AUG is directed to suitable position on the 30S ribosomal sub-unit by:

- (1) Shine-Dalgarno sequences
- (2) Consensus sequences
- (3) RNA polymerase
- (4) All of these

**107.** The equivalent of a structural gene is:

- (1) Operon
- (2) Recon
- (3) Muton
- (4) Cistron

- 108.** Which of the following rRNAs acts as structural RNA as well as ribozyme in bacteria?  
 (1) 23 S rRNA  
 (2) 5.8 S rRNA  
 (3) 5 S rRNA  
 (4) 18 S rRNA
- 109.** The wobble concept was proposed by  
 (1) Watson and Crick  
 (2) Nirenberg and Lederberg  
 (3) Nirenberg and Matthaei  
 (4) Crick
- 110.** The fruit fly *Drosophila melanogaster* was found to be very suitable for experimental verification of chromosomal theory of inheritance by Morgan and his colleagues because:  
 (1) It reproduces parthenogenetically  
 (2) A single mating produces two young flies  
 (3) Smaller female is easily recognisable from larger male  
 (4) It completes life cycle in about two weeks
- 111.** Which statements are not true regarding chromosome theory of inheritance?  
 (i) It was proposed by Sutton and Boveri indepently in 1902.  
 (ii) Close similarity between the transmission of hereditary characters and behaviour of chromosomes while passing from the one generation to the next through agency of gametes.  
 (iii) Both chromosomes as well as genes occur in pairs in the somatic or diploid cells.  
 (iv) A gamete contains only one chromosome of a type and only one of the two alleles of a character.  
 (v) The paired condition of both chromosomes as well as Mendelian factors is restored during fertilization.  
 (1) (i) and (iv)  
 (2) (iv) and (v)  
 (3) (ii) and (iii)  
 (4) None of these
- 112.** In a dihybrid cross, if you get 9 : 3 : 3 : 1 ratio it denotes that:  
 (1) The alleles of two genes are interacting with each other  
 (2) It is a multigentic inheritance  
 (3) It is a case of multiple allelism  
 (4) The alleles of two genes are segregating independently
- 113.** A homozygous organism is best described as which of the following  
 (1) dominant  
 (2) recessive  
 (3) hybrid  
 (4) true breed
- 114.** Percentage of recombination between A and B is 9%, A and C is 17% and B and C is 26%. The arrangement of genes would be:  
 (1) A–B–C  
 (2) A–C–B  
 (3) B–C–A  
 (4) B–A–C
- 115.** A test cross of  $F_1$  flies  $+a/+b$  produced the following offspring:  
 $++/ab = 9$   
 $ab/ab = 9$   
 $+b/ab = 41$   
 $a+/ab = 41$   
 What will be distance between linked gene?  
 (1) 82 cM  
 (2) 18 cM (*cis*)  
 (3) 20 cM  
 (4) 18 cM (*trans*)
- 116.** Genetic map is one that:  
 (1) Establishes sites of the genes on a chromosome  
 (2) Establishes the various stages in gene evolution  
 (3) Shows the stages during the cell division  
 (4) Shows the distribution of various species in a region

**117.** Which of the following statements is not true of two genes that show 50% recombination frequency?

- (1) The genes may be on different chromosomes
- (2) The genes are tightly linked
- (3) The genes show independent assortment
- (4) None of the above

**118.** In his classic experiments on pea plants, Mendel did not use:

- (1) Flower position
- (2) Seed colour
- (3) Pod length
- (4) Seed shape

**119.** Which one of the following generates new genetic combinations leading to variation?

- (1) Segregation of alleles
- (2) Independent assortment
- (3) Fertilisation
- (4) All of above

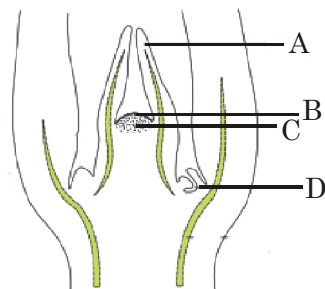
**120.** Read the statements (A–E) and answer the question following them.

- (A) Permanent tissues in plants do not have the capability of division.
- (B) After grazing by herbivores, grasses regenerate their lost parts by the action of intercalary meristem.
- (C) Interfascicular cambium is an example of apical meristem.
- (D) Metaxylem is a type of primary xylem.
- (E) Vessels are found in almost all seed plants.

How many statements are false?

- (1) 1
- (2) 2
- (3) 3
- (4) 4

**121.** Section of shoot apex is shown in the diagram. Select the option with correct identification of labelling and one of its features.



- (1) A–Plumule, gives rise to shoot
- (2) B–Shoot apex, possess apical meristem
- (3) C–Meristematic zone, cells are actively dividing giving rise to trichomes
- (4) D–Leaf primordium, give rise to leaves and lateral branches

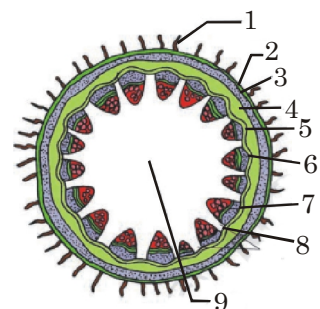
**122.** What is not common in tracheids and vessels?

- (1) Both are dead
- (2) Lignified cell walls
- (3) Structure
- (4) Absence of protoplasm

**123.** What is responsible for formation of annual rings?

- (1) Homogenous activity of cork cambium
- (2) Differential activity of cork cambium
- (3) Homogenous activity of vascular cambium
- (4) Differential activity of vascular cambium

**124.** In the given diagram which labelled parts are expected to be parenchymatous?



- (1) 2, 3, 6, 7
- (2) 4, 8, 9
- (3) 1, 4, 7, 9
- (4) 4, 8

**125.** Which of the following is not the part of stele?

- (1) Hypodermis
- (2) Pith
- (3) Secondary xylem
- (4) Vascular cambium

**126.** Starch sheath is found in:

- (1) Monocot leaf
- (2) Dicot leaf
- (3) Monocot root
- (4) Dicot stem

**127.** Casparian strips are modifications of.....in.....

- (1) Pericycle, dicot stem
- (2) Pericycle, dicot root
- (3) Endodermis, dicot root
- (4) Medullary rays, dicot stem

**128.** Leaves become fleshy to store food in:

- (1) Calotropis, Guava
- (2) Onion, Garlic
- (3) Turnip, Carrot
- (4) Colocasia, Zaminkand

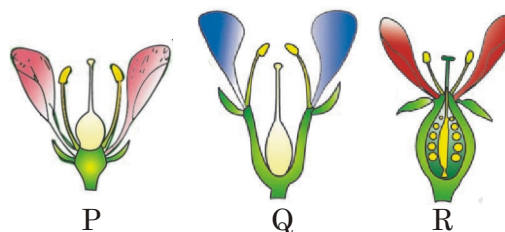
**129.** If 1-Salvia, 2-Brinjal, 3-Lily, 4-Pea, 5-China rose, 6-Citrus, then which option indicates the plants with stamens of variable length?

- (1) 1
- (2) 2, 4
- (3) 3, 5, 6
- (4) 1, 3, 4

**130.** Symmetry of Datura flowers is similar to that of:

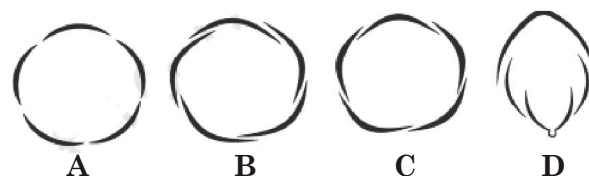
- (1) Chilli
- (2) Pea
- (3) Canna
- (4) Gulmohar

**131.** Three diagrams of flower indicating various positions of ovary with respect to other floral whorls are given. Which of the following option is correctly identifying them alongwith examples?



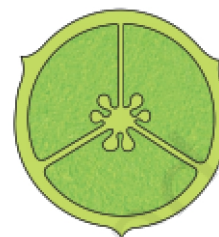
- (1) P–Hypogynous condition, Brinjal
- (2) Q–Epigynous condition, Peach
- (3) Q–Perigynous condition, China rose
- (4) R–Hypogynous condition, Guava

**132.** Four types of aestivations are given in the diagram (A–D). Select the option with correct identification of at least two of them (A–D)



- (1) A–Twisted, D–Vexillary
- (2) B–Valvate, C–Imbricate
- (3) A–Valvate, C–Imbricate
- (4) C–Vexillary, D–Imbricate

**133.** Select the option with suitable example in which the given placentation is found:



- (1) Sunflower
- (2) Pea
- (3) Primrose
- (4) Tomato

**134.** Coir in coconut is obtained from:

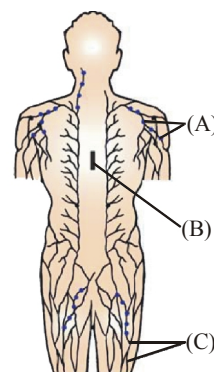
- (1) Epicarp
- (2) Mesocarp
- (3) Endocarp
- (4) Endosperm

**135.** How many copies of chromosome contained in each cell of many bacterial species in nearly all cases?

- (1) Two copies
- (2) 100 copies
- (3) Only one copy
- (4) Infinite copies

- 136.** Increased asthmatic attacks in certain seasons are due to
- (1) Inhalation of seasonal pollen
  - (2) Heat and humidity
  - (3) Low temperature
  - (4) Heavy rain fall
- 137.** True about plasma cells?
- (1) Are derived from T-cells
  - (2) Develop into B-cells
  - (3) Secrete large amount of Beta interferones.
  - (4) Specifically forms antibodies against particular pathogen.
- 138.** Antigen binding site in an antibody is found between
- (1) Two light chains
  - (2) Two heavy chains
  - (3) One heavy and one light chain
  - (4) Both 2 and 3 depending upon nature of antigen.
- 139.** Use of antihistamines and steroid give a quick relief from
- (1) Nausea
  - (2) Headache
  - (3) Allergy
  - (4) Cough.
- 140.** Common cold differs from pneumonia in
- (1) Pneumonia is communicable while common cold is deficiency disease.
  - (2) Pneumonia caused by genetic disorder while common cold by bacterial infection.
  - (3) Pneumonia is caused by a virus while common cold is caused by bacterium *Haemophilus influenzae*.
  - (4) Pneumonia infects alveoli while common cold affects nose and respiratory passage.
- 141.** Antivenom injection contains preformed antibodies while polio drops that are administered into the body contain
- (1) Gamma globulin
  - (2) Attenuated pathogens
  - (3) Activated pathogens
  - (4) Harvested antibodies

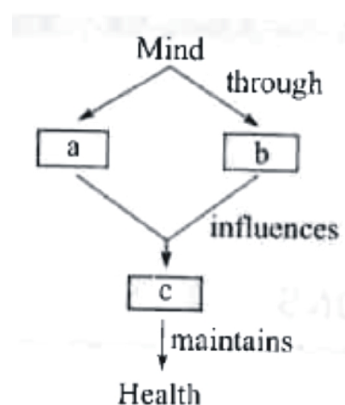
- 142.** In higher vertebrates, the immune system can distinguish self-cells and non-self. If this property is lost due to genetic abnormality and it attacks self-cells, then it leads to
- (1) Auto-immune disease
  - (2) Active immunity
  - (3) Allergic response
  - (4) Graft rejection
- 143.** In given diagramme identify (A), (B) and (C)



- (1) (A) – Blood vessels, (B) – Tonsils, (C) – Lymph nodes
  - (2) (A) – Lymphatic vessels, (B) – Bone marrow, (C) – Thymus
  - (3) (A) – Lymph nodes, (B) – Thymus, (C) – Lymphatic vessels
  - (4) (A) – Payer's patches, (B) – Spleen, (C) – Blood vessels
- 144.** Hepatitis-B vaccine is
- (1) First generation vaccine
  - (2) Second generation vaccine
  - (3) Third generation vaccine
  - (4) Fourth generation vaccine
- 145.** Bursa Fabricius of birds is associated with
- (1) Generation of basophils
  - (2) Generation of B-cells
  - (3) Metabolism of fatty acids
  - (4) Production of uric acid
- 146.** Best HLA match for transplants in order of preference is
- (1) Parent > Sibling > Twin > Unrelated donor
  - (2) Sibling > Twin > Parent > Unrelated donor
  - (3) Twin > Sibling > Parent > Unrelated donor
  - (4) Twin > Unrelated donor > Parent > Sibling

- 147.** A person is injected with gammaglobulin against hepatitis B. it is  
 (1) Artificially acquired active immunity  
 (2) Naturally acquired active immunity  
 (3) Artificially acquired passive immunity  
 (4) Naturally acquired passive immunity
- 148.** Every antibody is specific because?  
 (1) Specific type of bonds found in each antibody  
 (2) There are only five type of antigens stimulates to form only 5 types of antibodies (IgG, A, M, D, E)  
 (3)  $F_c$  segment is different for each antibody  
 (4) Paratope is specifically designed for each type of epitope in every antibody.
- 149.** Which of the following is an example of attenuated or altered live agents, used for active immunization?  
 (1) Typhoid bacilli  
 (2) Sabin polio virus  
 (3) Tetanus bacteria  
 (4) Hepatitis B virus subunits
- 150.** Cells involved in specific immune mechanism are?  
 (1) Erythrocyte  
 (2) Lymphocyte  
 (3) Eosinophills  
 (4) Thrombocytes
- 151.** Which of the following drugs are used in the treatment of allergy and quickly reduce the symptom of allergy?  
 (A) Antihistamine (B) Adrenaline  
 (C) Steroids (D) Chloroquine  
 (1) A only (2) A and B only  
 (3) A, B and C only (4) A, B, C and D
- 152.** Migration of phagocytes and other leucocytes towards the place of infection occurs through  
 (1) Thrombosis (2) Diapedesis  
 (3) Emboly (4) Phagocytosis

- 153.** Mark out the correct statement regarding the natural killer cells—  
 (a) They are a part of innate cellular barrier.  
 (b) NK cells release perforin during their interaction with the target cells.  
 (c) NK cells destroy both tumours and virally infected cells.  
 (d) NK cells are a type of lymphocytes.  
 (1) a, b, c, d (2) b, c, d only  
 (3) a only (4) a, b only
- 154.** Which of the following possess class II MHC antigens?  
 (1) Antigen presenting cells  
 (2) Erythrocytes  
 (3) Cardiac muscle cells  
 (4) Skeletal muscle cells
- 155.** Fill in the blanks:



- (1) a—neural system, b—endocrine system, c—immune system  
 (2) b—neural system, c—endocrine system, a—immune system  
 (3) a—neural system, c—endocrine system, b—immune system  
 (4) c—neural system, b—endocrine system, a—immune system
- 156.** Diseases which are easily transmitted from one person to another, are called  
 (1) Infectious diseases  
 (2) Non-infectious diseases  
 (3) Deficiency diseases  
 (4) Genetic disorders

- 157.** In immunoglobulin disulphides do not join  
(1) Two heavy chains  
(2) Light chain with heavy chain  
(3) Two light chains  
(4) All of the above
- 158.** Each immunoglobulin has two heavy chains and two light chains. The antigen binding site is found in  
(1) Variable region of heavy chain  
(2) Variable region of light chain  
(3) Constant region of light chain  
(4) Variable region of both heavy and light chain
- 159.** Which of the following two statements **correct** for kidney transplant ?  
(a) Immuno suppressant drugs used for long time inspite kidney properly transplanted  
(b) Graft rejection is due to cell mediated immunity  
(c) B-cells responsible for graft rejection  
(d) Acceptance and rejection of graft depends on specific interferones.  
(1) (a) and (b)                      (2) (b) and (d)  
(3) (c) and (d)                      (4) (a) and (d)
- 160.** In case of IDDM, why insulin not given Intravenously ?  
(1) It is not effective and glucose uptake not happens  
(2) Chances of Insulin shock  
(3) Insulin more effectively works in other routes  
(4) Intradermal is the preferred route
- 161.** Damage to thymus in a child would lead to  
(1) Loss of cell mediated immunity  
(2) Loss of antibody mediated immunity  
(3) A reduction in stem cell production  
(4) Reduction in haemoglobin content of blood
- 162.** Active natural immunity means  
(1) Resistance developed before disease  
(2) Resistance developed after disease  
(3) Increased heart beat  
(4) Increased flow of blood
- 163.** Obstacle to large scale transplantation of organs is  
(1) Insufficiency of organ donors  
(2) Immunological rejection of foreign bodies  
(3) Religious or ethnic considerations  
(4) Lack of effective surgical techniques
- 164.** Antigens are found  
(1) Inside cytoplasm  
(2) Inside nucleus  
(3) On nuclear envelope  
(4) On cell surface
- 165.** Memory-based acquired immunity evolved in  
(1) Higher vertebrates  
(2) Lower vertebrates  
(3) Lower invertebrates  
(4) All of the above
- 166.** Diseases are broadly grouped into infectious and non-infectious diseases. In the list given below identify the infectious diseases.  
(i) AIDS                                      (ii) Cancer  
(iii) Common cold                      (iv) Allergy  
(1) (i) and (iii)                      (2) (i) and (ii)  
(3) (i), (ii), (iii)                      (4) (i), (ii), (iii), (iv)
- 167.** At new place you suddenly starts sneezing and wheezing. The statement correctly correlate this condition  
(i) This is exaggerated response of the immune system  
(ii) Symptoms are watery eyes and running nose  
(iii) Due to the release of chemicals like adrenaline, steroids and histamine  
(iv) Antibodies produced to these are IgE type  
(v) Rheumatoid arthritis is an example  
(1) i, ii, iii, iv are true  
(2) i, iii, iv are true  
(3) i, ii, iv are true  
(4) All are correct except v
- 168.** Incorrect about spleen is -  
(1) Secondary lymphoid organ  
(2) Bean - shaped  
(3) Contains phagocytes only  
(4) Traps blood - borne micro organisms
- 169.** Which cells has phagocytic nature ?  
(A) PMNL - neutrophil  
(B) Monocyte  
(C) NK cells  
(1) A, B                                      (2) B  
(3) A, B, C                                      (4) B, C

**170.** Fill in the blanks:

1. The ...a... lymphoid organs provide the sites for interaction of lymphocytes with antigen, which then proliferate to become ...b...
  2. The thymus is a lobed organ located near the heart and beneath the ...C...
  3. Spleen has a large reservoir of ...d...
- (1) a–primary, b–antigen sensitive cells, c–collar bone, d–lymphocytes  
 (2) a–secondary, b–receptor cells, c–breastbone, d–erythrocytes  
 (3) a–primary, b–effector cells, c–breast bone, d–lymphocytes  
 (4) a–secondary, b–effector cells, c–breastbone, d–erythrocytes

**171.** Match the following

| Column-I |                 | Column-II |                                                    |
|----------|-----------------|-----------|----------------------------------------------------|
| A        | PCT             | i         | Selective secretion of $H^+$ and $K^+$ ions        |
| B        | DCT             | ii        | 70-80 percent of electrolytes and water reabsorbed |
| C        | Descending limb | iii       | Allow passage of small amount of urea              |
| D        | Ascending limb  | iv        | permeable to water                                 |
| E        | Collecting duct | v         | permeable to salts                                 |

- (1) A-i ; B-ii ; C-iii ; D-iv ; E-v  
 (2) A-i ; B-iii; C-iv ; D-ii ; E-v  
 (3) A-ii; B-i; C-iv; D-v; E-iii  
 (4) A-ii : B-i ; C-v ; D-iv ; E-iii

**172.** Which one of the following statements are correct :-

- (1) The medullary zone of kidney is divided into a few conical masses called medullary pyramids projecting into the calyces
- (2) Inside the kidney the cortical region extends in between the medullary pyramids
- (3) Glomerulus alongwith Bowman's capsule is called the renal corpuscle
- (4) All the above

**173.** Regulation of kidney function of Juxtaglomerular Apparatus (JGA) involves certain steps given below. Arrange them in the correct order :-

- (a) Release of enzyme renin
  - (b) Release of aldosterone from adrenal gland
  - (c) Reabsorption of  $Na^+$  and water at DCT
  - (d) Decrease in blood pressure and blood volume
  - (e) Conversion of angiotensinogen to angiotensin II
- (1) a, e, b, c, d                      (2) b, c, a, e, d  
 (3) d, a, b, e, c                      (4) d, a, e, b, c

**174.** Which of the following statements is incorrect?

- (1) Tadpole larva is ammonotelic but adult frog is ureotelic
- (2) Sharks and rays are uricotelic
- (3) Ammonia is the most toxic and most soluble waste
- (4) Animals that lay shelled eggs are uricotelic

**175.** Urea is synthesized from

- (1)  $CO_2$  and uric acid
- (2)  $CO_2$  and ammonia
- (3) Ammonia and uric acid
- (4)  $CO_2$  and water

**176.** Internal section of Kidney shows extension of cortex into medullary regions between the pyramids and is called

- (1) Ducts of Bellini                      (2) Major calyx
- (3) Minor calyx                          (4) Columns of Bertini

**177.** The protective layer of Kidney is/are:

- (1) Renal capsule                      (2) Adipose tissue
- (3) Renal fascia                          (4) All of these

**178.** The structural and functional unit of Kidney is called

- (1) Nephron
- (2) Uriniferous tubule
- (3) Both (1) and (2)
- (4) Malpighian tubule

**179.** The renal threshold for glucose is

- (1) 100 mg/dL                              (2) 120 mg/dL
- (3) 180 mg/dL                              (4) 250 mg/dL

**180.** Accumulation of urea in blood of patients suffering with renal failure is called

- (1) Renal calculi
- (2) Glomerulonephritis
- (3) Uremia
- (4) Ketosis

